



**INTERNATIONAL SOCIETY FOR ROCK MECHANICS
COMMISSION ON
STANDARDIZATION OF LABORATORY AND FIELD TESTS**

Suggested Methods for Determining Shear Strength

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**COMMITTEE ON FIELD TESTS
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PART 1. SUGGESTED METHOD FOR IN SITU DETERMINATION OF DIRECT SHEAR STRENGTH

SCOPE

1. (a) This test measures peak and residual direct shear strength as a function of stress normal to the sheared plane. Results are usually employed in limiting equilibrium analysis of slope stability problems or for the stability analysis of dam foundations [1-3*].

(b) The inclination of the test block and system of applied loads is usually selected so that the sheared plane coincides with a plane of weakness in the rock, for example a joint, plane of bedding, schistosity or cleavage, or with the interface between soil and rock or concrete and rock [4].

(c) A shear strength determination should preferably comprise at least five tests on the same test horizon

with each specimen tested at a different but constant normal stress.

(d) In applying the results of the test, the pore water pressure conditions and the possibility of progressive failure must be assessed for the design case as they may differ from the test conditions.

APPARATUS

2. (a) Equipment for cutting and encapsulating the test block, rock saws, drills, hammer and chisels, formwork of appropriate dimensions and rigidity, expanded polystyrene sheeting or weak filler, and materials for reinforced concrete encapsulation.

3. Equipment for applying the normal load (i.e. Fig. 1) including:

(a) Flat jacks, hydraulic rams or dead load of sufficient capacity to apply the required normal loads [5].

(b) A hydraulic pump if used should be capable of maintaining normal load to within 2% of a selected value throughout the test.

(c) A reaction system to transfer normal loads uniformly to the test block, including rollers or a similar

* Numbers refer to Notes at the end of the text.

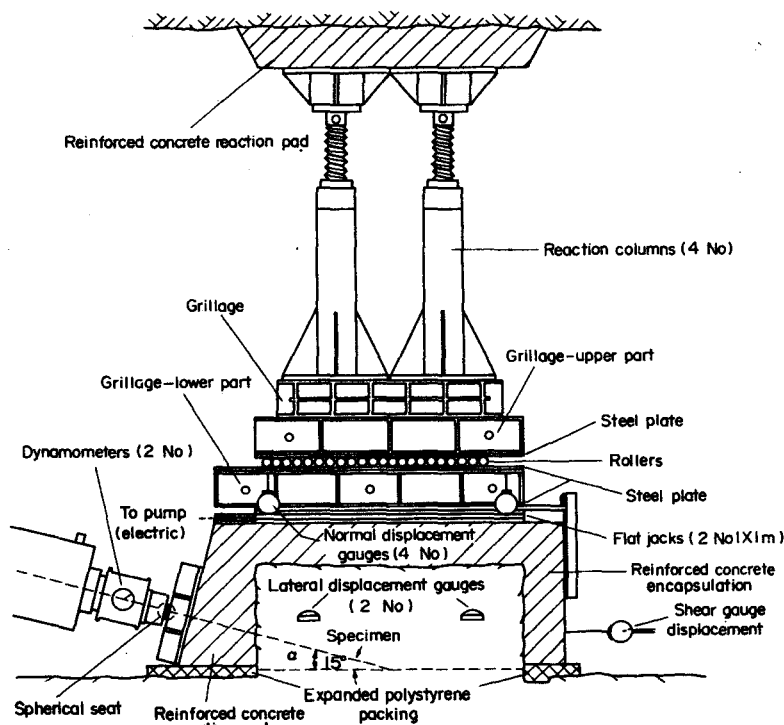


Fig. 1. Typical arrangement of equipment for *in situ* direct shear test.

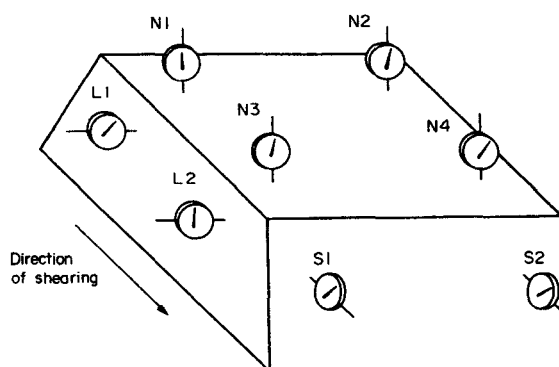


Fig. 2. Arrangement of displacement gauges: S1 and S2 for shear displacement, L1 and L2 for lateral displacement, N1-N4 for normal displacement.

low-friction device to ensure that at any given normal load the resistance to shear displacement is less than 1% of the maximum shear force applied in the test. Rock anchors, wire ties and turnbuckles are usually required to install and secure the equipment.

4. Equipment for applying the shear force (i.e. Fig. 1) including:

(a) One or more hydraulic rams [5] or flat jacks of adequate total capacity with at least 70 mm travel.

(b) A hydraulic pump to pressurize the shear force system.

(c) A reaction system to transmit the shear force to the test block. The shear force should be distributed uniformly along one face of the specimen. The resultant line of applied shear forces should pass through the centre of the base of the shear plane [6], with an angular tolerance of $\pm 5^\circ$.

5. Equipment for measuring the applied forces including:

(a) One system for measuring normal force and another for measuring applied shearing force with an accuracy better than $\pm 2\%$ of the maximum forces reached in the test. Load cells (dynamometers) or flat jack pressure measurements may be used. Recent calibration data applicable to the range of testing should be appended to the test report. If possible the gauges should be calibrated both before and after testing.

6. Equipment for measuring shear, normal and lateral displacements:

(a) Displacements should be measured (for example using micrometer dial gauges [7]) at eight locations on the specimen block or encapsulating material, as shown in Fig. 2.

(b) The shear displacement measuring system should have a travel of at least 70 mm and an accuracy better than 0.1 mm. The normal and lateral displacement measuring systems should have a travel of at least 20 mm and an accuracy better than 0.05 mm. The measuring reference system (beams, anchors and clamps) should, when assembled, be sufficiently rigid to meet these requirements. Resetting of gauges during the test should if possible be avoided.

PROCEDURE

7. Preparation:

(a) The test block is cut to the required dimensions (usually $700 \times 700 \times 350$ mm) using methods that avoid disturbance or loosening of the block [8,9]. The base of the test block should coincide with the plane to be sheared and the direction of shearing should correspond if possible to the direction of anticipated shearing in the full-scale structure to be analysed using the test results. The block and particularly the shear plane should unless otherwise specified be retained as close as possible to its natural *in situ* water content during preparation and testing, for example by covering with saturated cloth. A channel approximately 20 mm deep by 80 mm wide should be cut around the base of the block to allow freedom of shear and lateral displacements.

(b) A layer at least 20 mm thick of weak material (e.g. foamed polystyrene) is applied around the base of the test block, and the remainder of the block is then encapsulated in reinforced concrete or similar material of sufficient strength and rigidity to prevent collapse or significant distortion of the block during testing. The encapsulation formwork should be designed to ensure that the load-bearing faces of the encapsulated block are flat (tolerance ± 1 mm) and at the correct inclination to the shear plane (tolerance $\pm 2^\circ$).

(c) Reaction pads, anchors, etc., if required to carry the thrust from normal and shear load systems to adjacent sound rock, must be carefully positioned and aligned. All concrete must be allowed time to gain adequate strength prior to testing.

8. Consolidation:

(a) The consolidation stage of testing is to allow pore water pressures in the rock and filling material adjacent to the shear plane to dissipate under full normal stress before shearing. Behaviour of the specimen during consolidation may also impose a limit on permissible rate of shearing (see paragraph 9(c)).

(b) All displacement gauges are checked for rigidity, adequate travel and freedom of movement, and a preliminary set of load and displacement readings is recorded.

(c) Normal load is then raised to the full value specified for the test, recording the consequent normal displacements (consolidation) of the test block as a function of time and applied loads (Figs 3 and 4).

(d) The consolidation stage may be considered complete when the rate of change of normal displacement recorded at each of the four gauges is less than 0.05 mm in 10 minutes. Shear loading may then be applied.

9. Shearing:

(a) The purpose of shearing is to establish values for the peak and residual direct shear strengths of the test horizon [2]. Corrections to the applied normal load may be required to hold the normal stress constant; these are defined in paragraph 10(c).

(b) The shear force is applied either in increments or continuously in such a way as to control the rate of shear displacement.

DIRECT SHEAR TEST DATA SHEET

Client:		Project: Concrete Dam		Location: Alcántara		Loc. No.:		Block No.:	
TEST BLOCK SPECIFICATION		See drawings & photographs Nos.:							
General rock description, index properties and water conditions		Normal conditions							
Description and index properties of surface to be sheared		Type:							
Dip:	Dip direction:	Roughness:	Filling & alteration:	Initial area A: 0.490 m ²					
Persistence:	Spacing of set:	Surface dimensions: 0.70 x 0.70							

FORCES

$$P_n = P_{na} + P_{sa} \sin \alpha$$

$$P_s = P_{na} \cos \alpha$$

$$C_n = P_n / A$$

$$\tau = P_s / A$$

(+) (-) Normal displacement
 $\alpha = 15^\circ$ ϕ_n (nominal)

1 Time elapsed (min)	2 Applied normal force P_n		3 Normal displacement Δ_n					4 Applied shear force P_s		5 Shear displacement Δ_s		6 Contact area A (corrected) m ²	7 P_{na} (kN)	8 C_n (MPa)	9 P_{sa} (kN)	10 τ (MPa)
	Reading	Force (kN)	1	2	3	4	Average (mm)	Reading	Force (kN)	1	2					
10		196	0.100	0.070	0.130	0.070			0	0	0	-	0.490	196		0
35		233	0.130	0.085	0.140	0.090			137	0.05	0.05	0.05				142
48		270	0.050	0.065	0.285	0.290			275	0.55	0.35	0.45				284
64		306	-0.200	0.010	0.435	0.495			412	1.35	1.10	1.22				426
87		343	-0.710	-0.205	0.600	0.720			549	2.55	2.30	2.42				568
109		380	-1.165	-0.445	0.680	0.850			686	3.90	3.50	3.70				710
131		417	-1.675	-0.615	0.710	0.970			824	5.15	4.60	4.88				853
154		453	-1.965	-0.745	0.720	1.050			961	6.10	5.50	5.80				995
172		490	-2.245	-0.880	0.720	1.105			1098	7.20	6.50	6.85				1137
189		527	-2.480	-1.055	0.695	1.165			1235	8.20	7.40	7.80				1279
206		504	-2.750	-1.205	0.640	1.165			1373	9.45	8.45	8.95				1421
234		601	-3.075	-1.505	0.465	1.100			1510	11.00	10.00	10.50				1563
252		637	-3.350	-1.830	0.280	0.910			1647	12.45	11.40	11.92				1705
264		674	-3.675	-2.185	0.050	0.720			1784	14.00	12.80	13.40				1847
276		711	-4.005	-2.665	-0.290	0.360			1922	15.55	14.40	14.98				1989
289		748	-4.585	-3.125	-0.890	-0.020			2059	17.60	16.45	17.02				2132
293	Rupture	784	-4.975	-3.375	-1.250	-0.290			2196	20.00	19.55	19.78				2274

Calibration data

Remarks

Tested by:

Date:

Fig. 3. Example layout of direct shear test data sheet.

(c) Approximately 10 sets of readings should be taken before reaching peak strength (Figs 3 and 5). The rate of shear displacement should be less than 0.1 mm/min in the 10-minute period before taking a set of readings. This rate may be increased to not more than 0.5 mm/min between sets of readings provided that the peak strength itself is adequately recorded. For a 'drained' test, particularly when testing clay-filled dis-

continuities, the total time to reach peak strength should exceed $6 t_{100}$ as determined from the consolidation curve (see paragraph 10(a) and Fig. 4). If necessary the rate of shear should be reduced or the application of later shear force increments delayed to meet this requirement.

(d) After reaching peak strength, readings should be taken at increments of from 0.5 to 5 mm shear displacement as required to adequately define the force-displacement curves (Fig. 5). The rate of shear displacement should be 0.02–0.2 mm/min in the 10-minute

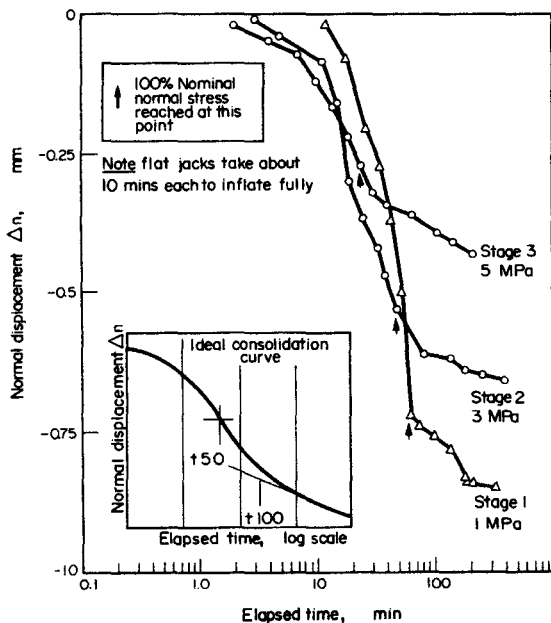
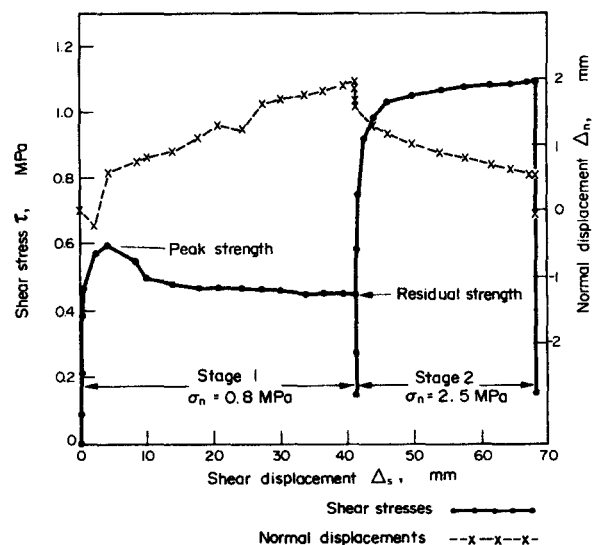
Fig. 4. Consolidation curves for a three-stage direct shear test, showing the construction used to estimate t_{100} .

Fig. 5. Shear stress-displacement graphs.

period before a set of readings is taken, and may be increased to not more than 1 mm/min between sets of readings.

(e) It may be possible to establish a residual strength value when the sample is sheared at constant normal stress and at least four consecutive sets of readings are obtained which show not more than 5% variation in shear stress over a shear displacement of 1 cm [11].

(f) Having established a residual strength the normal stress may be increased or reduced [12] and shearing continued to obtain additional residual strength values. The specimen should be reconsolidated under each new normal stress (see paragraph 8(d)) and shearing continued according to criteria given in paragraphs 9(c)–9(e) above.

(g) After the test the block should be inverted, photographed in colour and fully described (see paragraph 11). Measurements of the area, roughness, dip and dip direction of the sheared surface are required, and samples of rock, infilling and shear debris should be taken for index testing.

CALCULATIONS

10. (a) A consolidation curve (Fig. 4) is plotted during the consolidation stage of testing. The time t_{100} for completion of "primary consolidation" is determined by constructing tangents to the curve as shown. The time to reach peak strength from the start of shear loading should be greater than $6t_{100}$ to allow pore pressure dissipation [10].

(b) Displacement readings are averaged to obtain values of mean shear and normal displacements Δ_s and Δ_n . Lateral displacements are recorded only to evaluate specimen behaviour during the test, although if appreciable they should be taken into account when computing corrected contact area.

(c) Shear and normal stress are computed as follows:

$$\text{Shear stress } \tau = \frac{P_s}{A} = \frac{P_{sa} \cos \alpha}{A}$$

$$\text{Normal stress } \sigma_n = \frac{P_n}{A} = \frac{P_{na} + P_{sa} \sin \alpha}{A}$$

where

P_s = total shear force; P_n = total normal force,
 P_{sa} = applied shear force; P_{na} = applied normal force,
 α = inclination of the applied shear force to the shear plane (if $\alpha = 0$, $\cos \alpha = 1$ and $\sin \alpha = 0$),
 A = area of shear surface overlap (corrected to account for shear displacement).

If α is greater than zero the applied normal force should be reduced after each increase in shear force by an amount $P_{sa} \sin \alpha$ in order to maintain the normal stress approximately constant. The applied normal force may be further reduced during the test by an amount

$$\frac{\Delta_s(\text{mm}) \times P_n}{700} \text{ to compensate for area changes.}$$

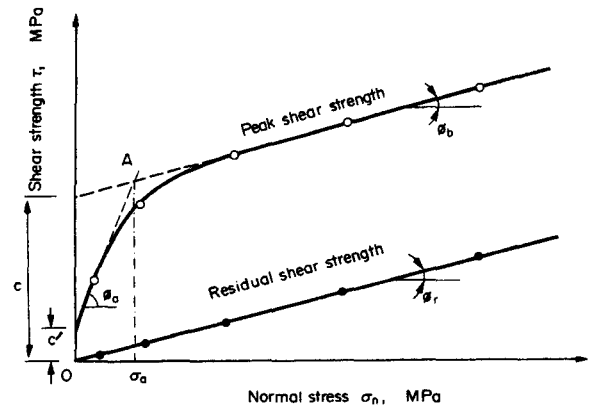


Fig. 6. Shear strength-normal stress graph.

(d) For each test specimen graphs of shear stress (or shear force) and normal displacement vs. shear displacement are plotted (Fig. 5), annotated to show the nominal normal stress and any changes in normal stress during shearing. Values of peak and residual shear strength and the normal stresses, shear and normal displacements at which these occur are abstracted from these graphs [2].

(e) Graphs of peak and residual shear strength vs. normal stress are plotted from the combined results for all test specimens. Shear strength parameters ϕ_a , ϕ_b , ϕ_r , c' and c are abstracted from these graphs as shown in Fig. 6.

ϕ_r = residual friction angle,

ϕ_a = apparent friction angle below stress σ_a ; point A is a break in the peak shear strength curve resulting from the shearing off of major irregularities on the shear surface. Between points 0 and A, ϕ_a will vary somewhat; measure at stress level of interest. Note also $\phi_a = \phi_u + i$, where ϕ_u is the friction angle obtained for smooth surfaces of rock on rock and angle i is the inclination of surface asperities.



ϕ_b = apparent friction angle above stress level σ_a (point A); note that ϕ_a will usually be equal to or slightly greater than ϕ_r and will vary somewhat with stress level; measure at the stress level of interest.

c' = cohesion intercept of peak shear strength curve; it may be zero.

c = apparent cohesion at a stress level corresponding to ϕ_b .

REPORTING OF RESULTS

11. The report should include the following:

(a) A diagram, photograph and detailed description of test equipment and a description of methods used for specimen preparation and testing. (Reference may be

made to this ISRM "suggested method" stating only departures from the prescribed techniques.)

(b) For each specimen a full geological description of the intact rock, sheared surface, filling and debris preferably accompanied by relevant index test data (e.g. roughness profiles; Atterberg limits, water content and grain size distribution of filling materials).

(c) Photographs of each sheared surface together with diagrams giving the location, dimensions, area, dip and dip direction and showing the directions of shearing and any peculiarities of the blocks.

(d) For each test block a set of data tables, a consolidation graph and graphs of shear stress and normal displacement vs. shear displacement (e.g. Figs 3, 4 and 5). Abstracted values of peak and residual shear strength should be tabulated with the corresponding values of normal stress, shear and normal displacement.

(e) For the shear strength determination as a whole, graphs and tabulated values of peak and residual shear strength vs. normal stress, together with derived values for the shear strength parameters (e.g. Fig. 6).

PART 2. SUGGESTED METHOD FOR LABORATORY DETERMINATION OF DIRECT SHEAR STRENGTH

SCOPE

1. (a) This test measures peak and residual direct shear strength as a function of stress normal to the sheared plane. Results are employed, for example, in the limiting equilibrium analysis of slope stability problems or for the stability analysis of dam foundations [1-3*].

(b) The inclination of the test specimen with respect to the rock mass, and its direction of mounting in the testing machine are usually selected so that the sheared plane coincides with a plane of weakness in the rock, for example a joint, plane of bedding, schistosity or cleavage, or with the interface between soil and rock or concrete and rock [4].

(c) A shear strength determination should preferably comprise at least five tests on the same test horizon with each specimen tested at a different but constant normal stress.

(d) In applying the results of the test, the pore-water pressure conditions and the possibility of progressive failure must be assessed for the design case as they may differ from the test conditions.

APPARATUS

2. Equipment for taking specimens of rock, including:

(a) Equipment for cutting the specimen; for example a large-diameter core drill, percussive drills, rock saws or hammers and chisels, also equipment for measuring the dip, dip direction, roughness and other characteristic features of the test horizon.

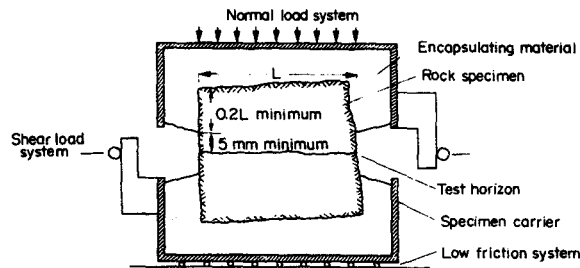


Fig. 7. Arrangement for laboratory direct shear test.

(b) Materials for holding the specimen together, for example binding wire or metal bands.

(c) Materials to protect the specimen against mechanical damage and changes in water content both during cutting and transit to the laboratory, for example protective packing and wax or similar waterproofing material.

3. Equipment for mounting the specimen including:

(a) Specimen carriers, forming a dismountable part of the test equipment.

(b) Cement, plaster, resin or similar strong encapsulating materials together with appropriate mixing utensils.

4. Testing equipment (a shear box, for example Fig. 7) incorporating:

(a) A means of applying the normal load, typically a hydraulic, pneumatic or dead-weight mechanical system, designed to ensure that the load is uniformly distributed over the plane to be tested. The resultant force should act normal to the shear plane, passing through its centre of area. The system should have a travel greater than the amount of dilation or consolidation to be expected, and should be capable of maintaining normal load to within 2% of a selected value throughout the test [5].

(b) A means of applying the shear force, typically a hydraulic jack or a mechanical gear-drive system, designed so that the load is distributed uniformly along one half-face of the specimen with the resultant applied shear force acting in the plane of shearing. The equipment should be designed for a shear travel greater than 10% of the specimen length. It should include rollers, cables or a similar low friction device to ensure that resistance of the equipment to shear displacement is less than 1% of the maximum shear force applied in the test.

(c) Equipment for independent measurement of the applied shear and normal forces, with an accuracy better than $\pm 2\%$ of the maximum forces reached in the test. Recent calibration data applicable to the range of testing should be appended to the test report.

(d) Equipment for measuring shear, normal and lateral displacement, for example micrometer dial gauges or electric transducers. Gauges may be mounted as shown in Fig. 2, or the four normal displacement gauges may be replaced by a single gauge mounted centrally. The shear displacement measuring system should have a travel greater than 10% of the specimen

* Numbers refer to Notes at the end of the text.

length and an accuracy better than 0.1 mm. The normal and lateral displacement measuring systems should have a travel greater than 20 mm and an accuracy better than 0.05 mm. Resetting of gauges during the test should if possible be avoided. If electric transducers or an automatic recording system is used a recent calibration should be included in the report.

PROCEDURE

5. Preparation:

(a) The test horizon is selected and dip, dip direction and other relevant geological characteristics are recorded. Block or core specimens containing the test horizon are collected using methods selected to minimize disturbance, if possible in such a way as to retain natural water content. The specimen dimensions and the location of the test horizon within the block or core should if possible allow mounting without further trimming in the laboratory, and with sufficient clearance for adequate encapsulation [8]. The test plane should preferably be square with a minimum area of 2500 mm² [2]. The mechanical integrity of the specimens should be preserved by binding tightly with wire or tape which is to be left in position until immediately before testing.

(b) Specimens that are not encapsulated immediately for testing should be given a waterproof coating, labelled and packaged to avoid damage in transit to the laboratory. Fragile specimens require special treatment, for example packaging in polyurethane foam (Stimpson, B., Metcalfe, F. G., and Walton, G., 1970. *Q. J. Engng Geol.* 3, No. 2, p. 127).

(c) The protective packaging, with the exception of the steel wire, is removed and the block supported in one of the carriers so that the horizon to be tested is secured in the correct position and orientation. The encapsulating material is poured and, after this has set, the other half-specimen is encapsulated in a similar manner. A zone at least 5 mm either side of the shear horizon should be free from encapsulating material.

6. Consolidation:

(a) The consolidation stage of testing is to allow pore water pressures in the rock and filling material adjacent to the shear plane to dissipate under full normal stress before shearing. Behaviour of the specimen during consolidation may also impose a limit on permissible rate of shearing (see paragraph 7(e)).

(b) Having mounted the specimen in the shear box, all gauges are checked and a preliminary set of load and displacement readings is recorded.

(c) Normal load is raised to the full value specified for the test, recording the consequent normal displacement (consolidation) of the specimen as a function of time and applied loads (Figs 3 and 4).

(d) The consolidation stage may be considered complete when the rate of change of normal displacement is less than 0.05 mm in 10 minutes. Shear loading may then be applied.

7. Shearing:

(a) The purpose of shearing is to establish values for

the peak and residual direct shear strengths of the test horizon [2].

(b) The shear force may be applied in increments but is usually applied continuously in such a way as to control the rate of shear displacement.

(c) Approximately 10 sets of readings should be taken before reaching peak strength (Figs 3 and 5). The rate of shear displacement should be less than 0.1 mm/min in the 10-minute period before taking a set of readings. This rate may be increased to not more than 0.5 mm/min between sets of readings provided that the peak strength itself is adequately recorded. For a "drained" test [10] particularly when testing clay-filled discontinuities, the total time to reach peak strength should exceed $6t_{100}$ as determined from the consolidation curve (see paragraph 8(a) and Fig. 4). If necessary the rate of shear should be reduced or the application of later shear force increments delayed to meet this requirement.

(d) After reaching peak strength, readings should be taken at increments of from 0.5 to 5 mm shear displacement as required to adequately define the force-displacement curves (Fig. 5). The rate of shear displacement should be 0.02–0.2 mm/min in the 10-minute period before a set of readings is taken, and may be increased to not more than 1 mm/min between sets of readings.

(e) It may be possible to establish a residual strength value when the sample is sheared at constant normal stress and at least four consecutive sets of readings are obtained which show not more than 5% variation in shear stress over a shear displacement of 1 cm [11].

(f) Having established a residual strength the normal stress may be increased or reduced [12] and shearing continued to obtain additional residual strength values. The specimen should be reconsolidated under each new normal stress (see paragraph 6), and shearing continued according to criteria given in paragraphs 7(c) to 7(e) above.

(g) After the test the shear plane should be exposed and fully described (see paragraph 9). The area of the shear surface is measured and photographs may also be required. Samples of rock, infilling and shear debris should be taken for index testing.

CALCULATIONS

8. (a) A consolidation curve (Fig. 4) is plotted during the consolidation stage of testing. The time t_{100} for completion of "primary consolidation" is determined by constructing tangents to the curve as shown. The time to reach peak strength from the start of shear loading should be greater than $6t_{100}$ to allow pore pressure dissipation [10].

(b) Displacement readings are averaged to obtain values of mean shear and normal displacements Δ_s and Δ_n . Lateral displacements are recorded only to evaluate specimen behaviour during the test, although if appreciable they should be taken into account when computing corrected contact area.

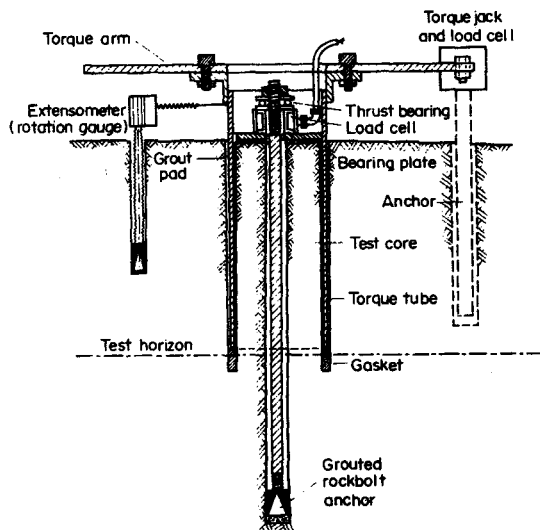


Fig. 8. Torsional shear test equipment.

(c) Shear and normal stress are computed as follows:

$$\text{Normal stress } \tau = \frac{P_s}{A}$$

$$\text{Shear stress } \sigma_n = \frac{P_n}{A}$$

where

P_s = total shear force; P_n = total normal force;

A = area of shear surface overlap (corrected to account for shear displacement).

(d) For each test specimen graphs of shear stress (or shear force) and normal displacement vs. shear displacement are plotted (Fig. 5), annotated to show the nominal normal stress and any changes in normal stress during shearing. Values of peak and residual shear strength and the normal stresses, shear and normal displacements at which these occur are abstracted from these graphs [2].

(e) Graphs of peak and residual shear strength vs. normal stress are plotted from the combined results for all test specimens. Shear strength parameters ϕ_a , ϕ_b , ϕ_r , c' and c are abstracted from these graphs as shown in Fig. 6.

REPORTING OF RESULTS

9. The report should include the following:

(a) A diagram and description of the test equipment and a description of the methods used for taking, packaging, transporting, storing, mounting and testing the specimen. (Reference may be made to this ISRM "suggested method" stating only departures from the prescribed techniques.)

(b) For each specimen a full geological description of the intact rock, sheared surface, filling and debris preferably accompanied by relevant index test data (e.g.

roughness profiles; Atterberg limits, water content and grain size distribution of filling materials).

(c) Diagrams and preferably photographs showing the sampling location, dip and dip direction of the tested horizon, also the dimensions and any peculiarities of the specimens.

(d) For each test block a set of data tables, a consolidation graph and graphs of shear stress and normal displacement vs. shear displacement (e.g. Figs 3, 4 and 5). Abstracted values of peak and residual shear strength should be tabulated with the corresponding values of normal stress, shear and normal displacement.

(e) For the shear strength determination as a whole, graphs and tabulated values of peak and residual shear strength vs. normal stress, together with derived values for the shear strength parameters (e.g. Fig. 6).

PART 3. SUGGESTED METHOD FOR IN SITU DETERMINATION OF SHEAR STRENGTH USING A TORSIONAL SHEAR TEST

SCOPE

1. (a) This test measures the shear strength of the base of a rock core which is rotated in its drillhole.

(b) The results may be used as an approximation to the direct shear strengths measured by the *in situ* or laboratory methods given in Parts 1 and 2 of this document [13*].

APPARATUS

2. Drilling equipment, including:

(a) A diamond core drill, water or air flush to suit the rock, capable of coring at diameters of approximately 50 mm and 300 mm to a depth greater than the core length required (usually 0.5–3 m).

3. Equipment for applying and measuring the normal load (Fig. 8) including:

(a) A rockbolt at least 1 m longer than the required core length, with grouted anchor at one end and a nut, thread and washer at the other.

(b) A bearing plate with external diameter equal to that of the test core, and a central hole allowing the plate to pass freely over the rockbolt.

(c) A hollow centre load cell fitting over the rockbolt, measuring tension in the bolt to $\pm 2\%$.

(d) A thrust bearing, concentric with the rockbolt and load cell, to prevent transmission of torque from the rockbolt to the core.

4. Equipment for applying and measuring the torque (Fig. 8) including:

(a) A torque tube to hold the rock core, fitting in the annular space around the core with an external clearance not less than 3 mm and an internal clearance sufficient to allow grouting between the tube and the core.

(b) A torque arm fixed rigidly across a diameter of the torque tube, perpendicular to the core axis and extending equally either side.

* Numbers refer to Notes at the end of the text.

(c) Two identical hydraulic jacks for applying the torque, pressurized by a single hydraulic pump, anchored rigidly to rockbolts equidistant from the core axis connected to the torque arm in such a way as to apply equal forces in opposite directions tangential to the core. Each jack is fitted with a load cell to measure to $\pm 2\%$ the applied force.

5. Equipment for measuring the rotation of the core, for example a dial gauge micrometer fixed to a rigid anchor and connected to the circumference of the torque tube by a wire and pulley system. The system should measure rotation within ± 0.001 radian and should have a travel greater than 0.1 radian.

PROCEDURE

6. Preparation:

(a) The test site is selected and a firm flat surface prepared perpendicular to the test horizon, usually 0.5–3 m above it. The test horizon is usually a plane of weakness in the rock such as a joint, plane of bedding or schistosity or the junction between concrete and rock.

(b) A small-diameter core (e.g. 50 mm) is drilled perpendicular to the anticipated test horizon and at least 1 m beyond it. This core is logged in detail, the test horizon selected and its depth measured to ± 10 mm. Additional small-diameter holes are drilled to take anchors for the torque jacks and rotation-measuring equipment.

(c) The normal load anchor is installed, with the anchor grout entirely below the test horizon. The bearing plate is installed over the bolt on a bed of grout, the load cell and thrust bearing placed in position and a normal load sufficient to rigidly hold the test core during drilling is applied.

(d) The test core is drilled to a depth $20 \text{ mm} \pm 5 \text{ mm}$ beneath the test horizon using the bearing plate to centralize the core barrel. The drilling method and drill fluid should be selected to minimize damage to the core.

(e) The annular space surrounding the test core is cleaned and filled with a weak material (e.g. sponge rubber gaskets) to a depth of 5–10 mm above the test horizon. The torque tube is greased on the outside, inserted to rest on the weak filler, and grouted to the core using resin or cement. The grout is left to gain strength and then the remaining apparatus is assembled and checked.

7. Testing:

(a) A selected normal load is applied by tightening the central rockbolt.

(b) The torque is gradually increased taking incremental readings of the torque and normal load cells and of the rotation gauge (e.g. Fig. 9). The rate of torque application should be such that 5–10 sets of readings are taken before reaching peak strength [2].

(c) Torsional loading is continued taking sets of readings at approximately 0.05 radian increments of rotation as shown in Fig. 9, until at least four sets of torque readings differing by less than 5% are obtained.

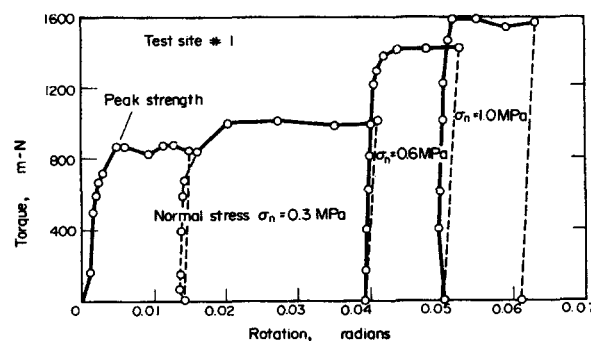


Fig. 9. Torque-rotation graph for the torsional shear test.

(d) The torsional loading is then removed, taking a further set of readings. The procedure 7(a) to 7(c) may then be repeated one or more times at successively greater normal loads depending on the maximum rotation permitted by the apparatus [12].

(e) The test core is removed and the sheared surface examined and described in detail. Samples of rock, filling material and shear debris may be taken for index testing.

CALCULATIONS

8. (a) Torque values are calculated using the load cell calibration curves to obtain the loads (kN) then multiplying by the torque radius (mm) to obtain the torque (m-N). The total torque T is the sum of the two applied torques.

(b) Normal loads are calculated from the load cell calibration curve, and converted into normal stresses σ_n using the equation:

$$\sigma_n = \frac{4P_n}{\pi(D^2 - d^2)}$$

where

P_n = applied normal load,

D = outer diameter of test core,

d = inner diameter of test core.

(c) Shear strengths τ corresponding to the peak and residual portions of the torque-displacement graph (Fig. 9, Notes 2 and 3) may be calculated using one of two assumptions:

(i) In strong, elastic rock assuming maximum stress at the outer circumference of the test core

$$\tau = \frac{0.5TD}{J}$$

(ii) In softer rock, assuming constant tangential stress at every radius across the failure plane

$$\tau = \frac{0.365TD}{J}$$

where

$$J = \frac{\pi(D^4 - d^4)}{32}$$

The latter assumption gives a more conservative estimate and should always be used for calculating residual shear strength, usually also for peak strength calculation.

(d) Graphs of peak and residual shear strength vs. normal stress are plotted from the combined results for all test specimens. Shear strength parameters may be abstracted from these graphs as shown in Fig. 6.

REPORTING OF RESULTS

9. The report should include the following:

(a) A diagram and detailed description of the test equipment and methods used for drilling and testing. (Reference may be made to this ISRM method stating only departures from the prescribed techniques.)

(b) For each test a full geological description of the test horizon, filling and shear debris preferably accompanied by relevant index test data.

(c) Photographs of the sheared surface together with diagrams giving the location, depth, dip and dip direction of the surface, the internal and external diameters and any peculiarities of the test core.

(d) For each test a set of data tables, graphs of shear stress and normal stress versus rotational displacement; abstracted values for peak and residual shear strengths and corresponding normal stresses with the assumptions made in their calculation.

(e) For the shear strength determination as a whole, graphs and tabulated values of peak and residual shear strength versus normal stress, together with derived values for the shear strength parameters (e.g. Fig. 6).

NOTES

1. Direct shear strength can be determined in the laboratory (using the method described in Part 2) if the plane to be tested is smooth and flat in comparison with the size of specimen, and if the specimen can be cut and transported without disturbance.

2. Definitions (clarified in Figs. 5 and 6):

Peak shear strength—the maximum shear stress in the complete shear stress-displacement curve.

Residual shear strength—the shear stress at which no further rise or fall in shear strength is observed with increasing shear displacement. A true residual strength may only be reached after considerably greater shear displacement than can be achieved in testing. The test value should be regarded as approximate and should be assessed in relation to the complete shear stress-displacement curve.

Shear strength parameters c and ϕ —respectively the intercept and angle to the normal stress axis of a tangent to the shear strength-normal stress curve at a normal stress that is relevant to design (see Fig. 6).

3. The measured peak strength can be applied directly to full-scale stability calculations only if the same type and size of roughness irregularities are present on the tested plane as on a larger scale. If this is not the case, the true peak strength should be obtained from the test data using appropriate calculations (for

example, Patton, F. D., 1966, *Proc. 1st Int. Cong. Rock Mech. ISRM*, Lisbon, Vol. 1, pp. 509–512; Ladanyi, B. and Archambault, G., 1970. In *Rock Mechanics—Theory and Practice* (W. H. Somerton, ed.), AIME, New York, pp. 105–125; Barton, N. R., 1971, *Proc. Symp. ISRM*, Nancy, Paper 1–8).

4. Tests on intact rock (free from planes of weakness) are usually carried out using laboratory triaxial testing. Intact rock can, however, be tested in direct shear if the rock is weak and if the specimen block encapsulation is sufficiently strong.

5. If a dead load is used for normal loading precautions are required to ensure accurate centring and stability. If two or more hydraulic rams are used either for normal or shear loading care is needed to ensure that they are identically matched and are in exact parallel alignment. Each ram should be provided with a spherical seat. The travel of rams and particularly of flat jacks should be sufficient to accommodate the full anticipated specimen displacement. A normal displacement of ± 5 –10 mm may be expected, depending on the clay content and roughness of the shear surface.

6. The applied shear force may act in the plane of shearing so that the angle α is 0 (Fig. 1). This requires a cantilever bearing member to carry the thrust from the shear jacks to the specimen. If a method is used where the shear force acts at some distance above the shear plane, the line of action of the shear jacks should be inclined to pass through the centre of area of the shear plane. The angle α for a specimen $700 \times 700 \times 350$ mm approximates to 15° depending on the thickness of encapsulation. Tests where both shear and normal forces are provided by a single set of jacks inclined at greater angles to the shear plane are not recommended, as it is then impossible to control shear and normal stresses independently.

7. The surface of encapsulating material is usually insufficiently smooth and flat to provide adequate reference for displacement gauges, and glass plates may be cemented to the specimen block for this purpose. These plates should be of adequate size to accommodate movement of the specimen. Alternatively a tensioned wire and pulley system with gauges remote from the specimen can be used. The system as a whole must be reliable and conform with specified accuracy requirements. Particular care is needed in this respect when employing electric transducers or automatic recording equipment.

8. A test block size of $700 \times 700 \times 350$ mm is suggested as standard for *in situ* testing. Smaller blocks may be permissible, for example, if the surface to be tested is relatively smooth; larger blocks may be needed when testing very irregular surfaces. The size and shape of the test block may for convenience be adjusted so that faces of the block coincide with natural joints or fissures; this minimizes block disturbance during preparation. Irregularities that would limit the thickness or emplacement of encapsulation material or reinforcement should be removed.

9. It is advisable particularly if the test horizon is

inclined at more than 10° – 20° to the horizontal to apply a small normal load to the upper face of the test specimen while the sides are cut, to prevent premature sliding and also to inhibit relaxation and swelling. The load, approximately 5–10% of that to be applied in the test, may, for example, be provided by screw props or a system of rock bolts and cross-beams, and should be maintained until the test equipment is in position.

10. The requirement that total time to reach peak strength should exceed $6 t_{100}$ is derived from conventional soil mechanics consolidation theory (for example, Gibson and Henkel, *Geotechnique* 4, pp. 10–11, 1954) assuming a requirement of 90% pore water pressure dissipation. This requirement is most important when testing a clay-filled discontinuity. In other cases it may be difficult to define t_{100} with any precision because a significant proportion of the observed "consolidation" may be due to rock creep and other mechanisms unrelated to pore pressure dissipation. Provided the rates of shear specified in the text are followed, the shear strength parameters may be regarded as having been measured under conditions of effective stress ("drained conditions").

11. An independent check on the residual friction

angle should be made by testing in the laboratory two prepared flat surfaces of the representative rock. The prepared surfaces should be saw-cut and then ground flat with No. 80 silicon carbide grit.

12. The normal load should when possible be applied in increasing rather than decreasing stages. Reversals of shear direction or resetting of the specimen block between normal load stages, sometimes used to allow a greater total shear displacement than would otherwise be possible, are not recommended because the shear surface is likely to be disturbed and subsequent results may be misleading. It is generally advisable, although more expensive, to use a different specimen block for each residual strength determination.

13. The torsional shear test is inexpensive and has the advantage of causing a minimum of disturbance to the test horizon. However, shear stress and displacement gradients generally exist from the inner to the outer radius of the core so that the shear strength values obtained must be regarded as approximate. The test should not be used for testing horizons that have a pronounced directional roughness. There is generally no provision for monitoring normal displacements or consolidation of the specimen.